

Algorithm to Display String Length

1. **Display prompt message:** Use the `sys_write` system call to prompt the user to enter a string.
2. **Accept string input:** Use the `sys_read` system call to read the user input into a buffer. The system call returns the number of bytes read (the string length) in the `rax` register.
3. **Store the length:** Store the length returned in `rax` to a dedicated memory variable.
4. **Convert length to ASCII string:** The length is an integer. To display it using `sys_write`, it must be converted into a string of ASCII characters. This is done by repeatedly dividing the number by 10 and converting each remainder to its ASCII character equivalent ('0' + remainder), storing them in reverse order in a temporary buffer.
5. **Display the length:** Use `sys_write` to display the "Length: " message and the calculated length (now in ASCII string format).
6. **Exit the program:** Use the `sys_exit` system call to terminate the program